

3 June 2017

Hedin equations

References:

- 1) L. Hedin, Phys. Rev. 139 A746 (1965)
- 2) F. Giustino, Rev. Mod. Phys. 89, 015003 (2017)
- 3) A. Stan, "Conserving approximations in nonequilibrium Green's function theory" PhD thesis Univ Groningen 2009.
- 4) Ferdi Aryasetawan "Functional derivative technique"

Introduction: The Feynman diagrammatic method was developed in the 1950 - 1960s which is intuitive and relatively simple. But the infinite set of diagrams are difficult to keep track. It is easy to miss a diagram or give incorrect prefactor (combinatorial factors).

Hedin came up in 1965 with a closed set of equations which is equivalent to the original diagrams. Another important feature of the Hedin equations is that it is expressed in the full Green's functions G not G_0 . This makes it easy to do approximations, such as GW method. Hedin's equations are closed only formally, because of the presence of functional derivatives, such as $\frac{\delta \Sigma}{\delta G}$.

Here I try to reproduce standard literature results for pure electron systems (ions are not moving), and the electron + phonon (without electron-electron interaction). For the most general case of electron, phonon, and full electron-electron, electron-phonon interacting system, I refer to the Rev Mod Phys long review by Giustino.

I still prefer a discrete description of electrons with a tight-binding model to start with

$$c^\dagger H c = \sum_{j,k} c_j^\dagger H_{j,k} c_k$$

where $c = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_j \end{pmatrix}$ is the annihilation operators vector, c_j for

site j (or state j). The charge at site j

is given by $\hat{q}_j = (-e) c_j^\dagger c_j = (-e) n_j$. $-e$ is electron charge

We ignore spins, or if we wish, we can include the spin degrees of freedom in the index j . To describe the electron Coulomb interaction, we can introduce a scalar field ϕ so that the interaction energy is

$$\sum_j \hat{q}_j \phi_j = \sum_j (-e) c_j^\dagger c_j \phi_j$$

However, ϕ will be viewed as only an external one, we write down the Coulomb interaction explicitly rather than mediated by field ϕ .

$$\text{so } \frac{1}{2} \sum_{j,k} \frac{1}{4\pi\epsilon_0 |\vec{r}_j - \vec{r}_k|} \hat{q}_j \hat{q}_k = \frac{1}{2} \sum_{j,k} \hat{q}_j V_{j,k} \hat{q}_k$$

In the summation $\sum_{j,k} = \sum_{j=1}^N \sum_{k=1}^N$ so both j,k and k,j is counted so we need to divide by 2

$$H_{\text{int}} = \frac{1}{2} q^T V q \quad q = \begin{pmatrix} q_1 \\ q_2 \\ \vdots \\ q_j \\ \vdots \\ q_N \end{pmatrix}$$

as a matrix multiplication

$$V = (V_{j,k}) = (V(\vec{r}_j, \vec{r}_k))$$

We see one problem since $V(\vec{r}=0) \rightarrow \infty$ the terms with $j=k$, the self interaction terms, gives infinity, we formally subtract off this contribution

$$H_{\text{int}} = \frac{1}{2} q^T V q = \frac{1}{2} \sum_{j=1}^N \sum_{k=1}^N V_{j,k} q_j q_k$$

$$\rightarrow \frac{1}{2} \sum_{j \neq k} V_{j,k} q_j q_k = \frac{1}{2} \sum_{j,k} V_{j,k} q_j q_k - \frac{1}{2} \sum_j V_{j,j} q_j q_j$$

$$= \frac{1}{2} \sum_{\substack{j,k \\ \text{over all } j, \\ \text{all } k}} (V_{j,k} q_j q_k - \delta_{j,k} V_{j,k} q_j q_k)$$

sum over all j ,
all k

$$V = \frac{1}{4\pi\epsilon_0 r} \quad \text{in 3D}$$

clearly $V_{j,k} = V_{k,j}$

$$= \frac{1}{2} \sum_{j,k} V_{j,k} (q_j q_k - \delta_{j,k} q_j q_k) = \frac{1}{2} \sum_{j,k} V_{j,k} (q_j q_k - \delta_{j,k} q_j^2)$$

$$= \frac{1}{2} e^2 \sum_{j,k} V_{j,k} [c_j^\dagger c_j c_k^\dagger c_k - \delta_{j,k} (c_j^\dagger c_j)^2]$$

clearly $(c_j^\dagger c_j)^2 = c_j^\dagger c_j$

Since the occupation number $n_j = c_j^\dagger c_j$ has eigen values of 0 or 1 only so $n_j (1 - n_j) = 0$

$$n_j = n_j^2$$

We find $H_{int} = \frac{1}{2} e^2 \sum_{jk} v_{jk} [\underbrace{c_j^\dagger c_j c_k^\dagger c_k}_{\text{cancel}} - \delta_{jk} c_j^\dagger c_k]$

since $c_j c_k^\dagger + c_k^\dagger c_j = \delta_{jk}$

$$H_{int} = \frac{1}{2} e^2 \sum_{jk} v_{jk} [c_j^\dagger (-c_k^\dagger c_j + \delta_{jk}) c_k - \delta_{jk} c_j^\dagger c_k]$$

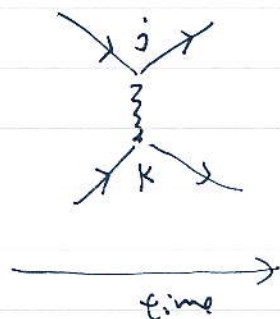
$$= \frac{1}{2} e^2 \sum_{jk} v_{jk} [-c_j^\dagger c_k^\dagger c_j c_k] \quad c_j c_k = -c_k c_j$$

$$= \frac{1}{2} e^2 \sum_{jk} v_{jk} c_j^\dagger c_k^\dagger c_k c_j = \frac{1}{2} e^2 \sum_{jk} c_j^\dagger c_k^\dagger v_{jk} c_k c_j$$

this is a standard form, usually written in the field $\psi(x)$ as

$$\frac{1}{2} e^2 \int dx dx' \psi^\dagger(x) \psi^\dagger(x') v(x, x') \psi(x') \psi(x)$$

We express the Coulomb interaction term in diagram as



we annihilate state j and k and create state k and j with amplitude v_{jk} . (order matters).

Putting together, our Hamiltonian is

$$\tilde{H} = c^\dagger H c + \sum_j \hat{q}_j \phi_j + \frac{1}{2} e^2 \sum_{jk} c_j^\dagger c_k^\dagger v_{jk} c_k c_j$$

$$\hat{q}_j = (-e) c_j^\dagger c_j$$

We can combine the 1st term and second term, $H \rightarrow H_{jk}^\phi = H_{jk} + (-e) \delta_{jk} \phi_j$

ϕ acts only on the diagonal (onsite). We will make ϕ time dependent $\rightarrow \phi_j(t)$.

The purpose of $\phi_j(t)$ is to be able to make functional derivative. Consider an arbitrary functional of $\phi_j(t)$

$$F(\phi_j(t)). \quad \delta F = F(\phi_j(t) + \delta\phi_j) - F(\phi_j(t))$$

$$= \sum_j \int dt \frac{\delta F}{\delta \phi_j(t)} \delta \phi_j(t)$$

This defines the functional derivative. In fact we'll make time $t \rightarrow \tau$ on the contour.

We define electron Green's function, contour ordered

$$G_{jk}(\tau, \tau') = -\frac{i}{\hbar} \langle T_\tau c_j(\tau) c_k^\dagger(\tau') \rangle_{\hat{H}}$$

Here we have the full Hamiltonian $\hat{H} = \hat{H}_0 + H_{int}$ which is time dependent due to the presence of ϕ .

G is a functional of ϕ , thus we can compute

$$\delta G / \delta \phi_j(\tau).$$

We follow the usual diagrammatic method with H_{int} as perturbation and $\hat{H}_0$ as the origin G_0 , then we can write the result in a Dyson equation

$$G = G^0 + G^0 M G$$

We use M as self energy as we are going to define the self energy by split M into a "Hartree" term and "Fock" term, and call only the Fock term Σ .

$$M = \text{Hartree} + \Sigma$$

$$= (-i\hbar) e^2 \delta(\tau, \tau') V G(\tau, \tau') + \Sigma \quad \begin{matrix} \text{need to be} \\ \downarrow \\ \text{disgr.} \end{matrix} = (-e) \delta(\tau, \tau') (V \langle q \rangle)$$

$$G_j(\tau, \tau^+) = \frac{i}{\hbar} \langle c_j^\dagger(\tau) c_j(\tau) \rangle = -\frac{1}{i\hbar} \langle q_j \rangle \left(\frac{1}{-e} \right)$$

Formally

$$G = G^0 + G^0 M G$$

$$\rightarrow G^0 G = I + M G$$

$$M = V + \Sigma$$

$$= I + (V + \Sigma) G$$

$$G^0 G^0 = I$$

$$\rightarrow (G^0 - V) G = I + \Sigma G$$

here $I \rightarrow \delta_{ij} \delta(\tau, \tau')$ Multiplication implies convolution in τ
 ie. $\Sigma G \rightarrow \int \Sigma_{ij}(\tau, \tau_1) G(\tau_1, \tau') d\tau_1$
 G^0 is the Green's function for H^0

$$\text{so } \left[i\hbar \frac{\partial}{\partial \tau} I - H^0 - (-e) \{ \Phi_j(t) \} \right] G^0 = I$$

so this is formally G^0^{-1} (except it is not in two variable convolution form)

$$\{ \Phi_j \} = \begin{bmatrix} \Phi_1 & \Phi_2 & 0 \\ 0 & & \Phi_n \end{bmatrix} \text{ is diagonal matrix with elements } \Phi_j$$

Combine with V term from the crystal self energy M
 we get the equation for G

$$V_{\text{tot}} = \{ V_j^{\text{tot}} \}$$

$$\left(i\hbar \frac{\partial}{\partial \tau} - H^0 - V_{\text{tot}}(-e) \right) G = I + \Sigma G$$

$$\begin{array}{c} \uparrow \\ \text{total} \\ \text{potential} \end{array} V_j^{\text{tot}} = \begin{array}{c} \uparrow \\ \text{extel} \\ \text{potential} \end{array} \Phi_j + \begin{array}{c} \text{induced potential} \\ \rightarrow \frac{1}{4\pi\epsilon_0} \int \frac{\rho(r')}{|r-r'|} dV \end{array}$$

V now change to potential

This is just a rewritten of the Dyson equation. The only unusual points are

- 1) we have an extel, time dependent potential
- 2) we split the Hartree term out and put to the left side of equation.
- 3) The Hartree term is $V G = D_0 G$, not $W G = D G_0$

The next step is to work out the equation of motion for the Green's function $G(\tau, \tau')$. As is well-known, for the linear system, we

have simple result. If $\hat{H} = c^\dagger H c$, then the counter ordered Green's function satisfies the "Schrödinger" equation $(i\hbar \frac{\partial}{\partial \tau} - \hat{H}) G(\tau, \tau') = \delta(\tau, \tau') \cdot I$ with a right-hand side δ -source, note the equation is still correct if the Hamiltonian is time dependent, $H \rightarrow H(\tau)$. The meaning of the time derivatives (and integrations) is explained in my 2nd review on NEGF in "Frontier Physics".

For interacting systems such as the Coulomb interactions $c^\dagger c^\dagger c c$, the equation of motion is obviously not closed. High order Green's functions involving four c 's result.

A key step and concept is the functional derivative with respect to g . With this mathematical device, the Coulomb interaction terms, the extra nonlinear term can be expressed as $\frac{\delta G}{\delta g(\tau)}$, as a result a "closed" equation is obtained for G . The rest of the steps are still complicated but the idea is to remove the reference to g from the problem, and make a system consistent of G , Σ , and vertex function Γ (at $g=0$) but still involves functional derivatives of Σ with respect to G .

First the equation of motion for G

$$G_{jk}(\tau, \tau') = -\frac{i}{\hbar} \langle T_\tau c_j(\tau) c_k^\dagger(\tau') \rangle$$

$$= \frac{1}{i\hbar} \theta(\tau, \tau') \langle c_j(\tau) c_k^\dagger(\tau') \rangle - \frac{1}{i\hbar} \theta(\tau', \tau) \langle c_k^\dagger(\tau') c_j(\tau) \rangle$$

Take derivative $i\hbar \frac{\partial}{\partial \tau}$, we note $\frac{\partial}{\partial \tau} \theta(\tau, \tau') = \delta(\tau, \tau')$

$\frac{\partial}{\partial \tau} \theta(\tau', \tau) = -\delta(\tau, \tau')$ and due to the δ -function

we need $\sum_j c_j(\tau) c_k^\dagger(\tau) + c_k^\dagger(\tau) c_j(\tau) = \delta_{jk}$ (ie $\tau' = \tau$)

this gives the right-hand side as $\delta(\tau, \tau') I$.

$I = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$ is $N \times N$ identity matrix. So we can

write $i\hbar \frac{\partial}{\partial \tau} G(\tau, \tau') = \delta(\tau, \tau') I + \left(\frac{1}{i\hbar}\right) \langle T_\tau \dot{c}(\tau) c^\dagger(\tau') \rangle$

Note $\langle \rangle$ means $\text{Tr}(\rho_H \dots)$ the time dependence of

c is the Heisenberg evolution of the full Hamiltonian

$$\hat{H} = H^\phi + H_{\text{int}} \quad \dot{c} = i[H, c] + \sum_j (-e) \hat{q}_j \dot{\varphi}_j(t) + H_{\text{int}}$$

Here $\dot{c}$ is $i\hbar \frac{dc}{d\tau} = [c, \hat{H}]$

$$\text{or } \dot{c}_j = \frac{dc_j}{d\tau} = \frac{1}{i\hbar} [c_j, c^\dagger [H^\phi] c] + \frac{1}{2} e^2 \sum_{k,l} c_k^\dagger c_l V_{kl} c_l c_k$$

Here H^ϕ is the original matrix H plus a diagonal term $(-e) \dot{\varphi}_j(t)$ on the (j,j) diagonal

$$\text{so } \dot{c}_j = \frac{1}{i\hbar} \left\{ \sum_k H_{jk}^\phi c_k + \frac{1}{2} e^2 [c_j, \sum_{k,l} c_k^\dagger c_l V_{kl} c_l c_k] \right\}$$

We have used the formula for commutator

$$[A, BC] = \{A, B\}C - B\{A, C\} = [A, B]C + B[A, C]$$

$$\{A, B\} = AB + BA$$

$$[A, B] = AB - BA$$

The Coulomb term gives $[c_j, c_k^\dagger c_l^\dagger c_l c_k]$

$$= [c_j, c_k^\dagger c_l^\dagger] c_l c_k + c_k^\dagger c_l^\dagger \underbrace{[c_j, c_l c_k]}_{=0}$$

$$= \{C_j, C_k^\dagger\} C_l^\dagger C_l C_k = C_k^\dagger \{C_j, C_l^\dagger\} C_l C_k$$

$$= \delta_{jk} C_l^\dagger C_l C_k - \underbrace{\delta_{jl} C_k^\dagger C_l C_k}_{\text{swap}} = \delta_{jk} n_l C_k + \delta_{jl} n_k C_l$$

hence $n_j = C_j^\dagger C_j$

Putting these result into the Heisenberg equation for C_j

we get $i\hbar \dot{C}_j = \sum_K H_{jk}^\phi C_K + \frac{1}{2} e^2 \sum_{K,l} (\delta_{jk} n_l C_K + \delta_{jl} n_K C_l) V_{Kl}$

Since n and C do not commute, the order in C can not be switched!

$$+ \frac{1}{2} e^2 \left[\sum_l n_l C_j V_{jl} + \sum_K n_K C_j V_{Kj} \right]$$

$$= \sum_K H_{jk}^\phi C_K + e^2 \left(\sum_l V_{jl} n_l \right) C_j \quad V_{jl} = V_{lj}$$

The physical meaning the 2nd term is clear. It is just the potential energy produced by all the charges located at l , in continuous form

$$V(\vec{r}) = \int \frac{e^2 n(\vec{r}')}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|} d\vec{r}'$$

Thus, the equation of motion for the canon ordered Green's function is

$$i\hbar \frac{\partial}{\partial \tau} G_{jk}(\tau, \tau') = \delta(\tau, \tau') \delta_{jk} + \left(\frac{1}{i\hbar} \right) \langle T_\tau \left[\sum_s H_{js}^\phi C_s(\tau) + e^2 \left(\sum_l V_{jl} n_l(\tau) \right) C_j(\tau) \right] C_k^\dagger(\tau') \rangle$$

$$= \delta(\tau, \tau') \delta_{jk} + \sum_s H_{js}^\phi G_{sk}(\tau, \tau') + \frac{e^2}{i\hbar} \sum_l V_{jl} \langle T_\tau n_l(\tau) C_j(\tau) C_k^\dagger(\tau') \rangle$$

We see a new special Green's function defined by

$$\left(-\frac{i}{\hbar} \right) \langle T \underbrace{C_l^\dagger(\tau) C_l(\tau)}_{n_l(\tau)} \underbrace{C_j(\tau) C_k^\dagger(\tau')}_{\text{}} \rangle$$

As far as time ordered is concerned the operator $n_l C$ must make as a single identity.

Now the critical step is to express the last nonlinear term as a functional derivative $\frac{\delta G}{\delta g}$. In order to work out this functional derivative explicitly, we need to have an explicit expression for G as a functional of g . To this end, we should use an interaction picture in terms of $\hat{H} = c^\dagger H c$ (not $\hat{H} g$). Another point is to note that g is defined for each site j and each given center time τ . Since each τ is independent (but we take only continuous function in τ) there is no reason to demand τ is a function of time t only. So $\tau \equiv (t, \sigma)$ is branch σ -dependent. As a result, the contribution to the scattering matrix from upper branch does not cancel the lower branch. i.e.

$$\text{Tr}(\rho(-\infty) T_\tau e^{-\frac{i}{\hbar} \int_C \tilde{H}_{\text{int}}(\tau) d\tau}) \neq 1 \quad \text{so we need}$$

to divide explicitly this term in the denominator. i.e. when g is center dependent, the vacuum diagrams are not zero, just like in the problem of full counting statistics.

But at the end of calculation, when the functional derivatives are obtained, we can set $g \equiv 0$. In this sense, the g -dependences are always treated in the linear response regime.

With the above discussion we have g -dependent Green's function

$$G_{jk}^g(\tau, \tau') \equiv (-\frac{i}{\hbar}) \frac{\text{Tr}[\rho(-\infty) T_\tau c_j(\tau) c_k(\tau') e^{-\frac{i}{\hbar} \int_{\text{int}} \tilde{H}(\tau') d\tau'}]}{\text{Tr}[\rho(-\infty) T_\tau e^{-\frac{i}{\hbar} \int_{\text{int}} \tilde{H}_{\text{int}}(\tau') d\tau'}]}$$

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Here $c(\tau)$ is in the interaction picture governed by $\hat{H} = c^\dagger H c$, i.e.

$$c(t) = e^{+\frac{i}{\hbar} \hat{H} t} c e^{-\frac{i}{\hbar} \hat{H} t}$$

$$\tilde{H}_{int}(\tau) = \tilde{H}_{int}^\varphi = (-e) \sum_{\mathbf{q}} \hat{n}_{\mathbf{q}}(\tau) \varphi_{\mathbf{q}}(\tau) + \sum_{\mathbf{q}} c^\dagger(\tau) M_{\mathbf{q}}^{\mathbf{q}} c(\tau) u(\tau)$$

$\hat{n}_{\mathbf{q}} = c_{\mathbf{q}}^\dagger c_{\mathbf{q}}$ the τ -dependence here is also in the interaction picture of $\hat{H}$. $\rho(-\infty)$ is some initial density matrix whose nature need not to be special here. The integral $\int d\tau''$ is on the Keldysh contour



The contour ordering superoperator applies to the operators to its right. Since $c(\tau)$ has no φ dependence, we have made explicit the φ -dependence in only the scattering operator S

$$S(\tau, \tau_2) = T_\tau e^{-\frac{i}{\hbar} \int_{\tau_2}^{\tau} d\tau'' [(-e) \sum_{\mathbf{q}} \hat{n}_{\mathbf{q}}(\tau'') \varphi_{\mathbf{q}}(\tau'') + c^\dagger M_{\mathbf{q}}^{\mathbf{q}} c u]} \quad \text{all}$$

$$G^\varphi(\tau, \tau') = \left(-\frac{i}{\hbar}\right) \frac{\text{Tr}[\rho(-\infty) T_\tau c(\tau) c^\dagger(\tau') S(-\infty, -\infty)]}{\text{Tr}[\rho(-\infty) S(-\infty, -\infty)]}$$

Note that $S(-\infty, -\infty) = S(-\infty, +\infty) S(+\infty, -\infty) \neq 1$
↓ lower branch upper branch

We make variation with respect to φ

$$\delta G^\varphi = \left(-\frac{i}{\hbar}\right) \left[\frac{\text{Tr}[\rho T_\tau c(\tau) c^\dagger(\tau') \delta S]}{\text{Tr}[\rho S]} - \frac{\text{Tr}[\rho T_\tau c(\tau) c^\dagger(\tau') S]}{(\text{Tr}[\rho S])^2} \times \text{Tr}[\rho \delta S] \right]$$

The variation of the scattering matrix with respect to φ is

$$\begin{aligned}
\delta S &= T_\tau \delta \left(e^{-\frac{i}{\hbar} \int d\tau'' \left[(-e) \sum_r n_r(\tau'') \phi_r(\tau'') + \dots \right]} \right) \\
&= T_\tau \left[-\frac{i}{\hbar} \int d\tau_1 (-e) \sum_r n_r(\tau_1) \delta \phi_r(\tau_1) e^{-\frac{i}{\hbar} \int d\tau'' \tilde{H}_{int}^\phi(\tau'')} \right] \\
&= \int d\tau_1 \frac{\delta S}{\delta \phi_r(\tau_1)} \delta \phi_r(\tau_1)
\end{aligned}$$

Thus we can identify the functional derivative of S with respect to ϕ as

$$\frac{\delta S}{\delta \phi_r(\tau_1)} = T_\tau \left[\left(-\frac{i}{\hbar} \right) (-e) n_r(\tau_1) e^{-\frac{i}{\hbar} \int d\tau'' \tilde{H}_{int}^\phi(\tau'')} \right]$$

Since δS is linear in $\delta \phi_r$ it is clear δG^ϕ is also linear in $\delta \phi_r$. We have

$$\begin{aligned}
\frac{\delta G_{jk}^\phi(\tau, \tau')}{\delta \phi_r(\tau_1)} &= \left(-\frac{i}{\hbar} \right) \left[\frac{\text{Tr}[\rho T_\tau c_j(\tau) c_k^\dagger(\tau') (-\frac{i}{\hbar}) (-e) n_r(\tau_1) S]}{\text{Tr}[\rho S]} \right. \\
&\quad \left. - \frac{\text{Tr}[\rho T_\tau c_j(\tau) c_k^\dagger(\tau') S]}{\text{Tr}[\rho S]} \cdot \frac{\text{Tr}[\rho T_\tau (-\frac{i}{\hbar}) (-e) n_r(\tau_1) S]}{\text{Tr}[\rho S]} \right]
\end{aligned}$$

$$= \left(\frac{1}{i\hbar} \right) \left\{ \left\langle T_\tau n_r(\tau_1) c_j(\tau) c_k^\dagger(\tau') \right\rangle_{\tilde{H}_{int}^\phi} \frac{ie}{\hbar} - \left(\frac{-e}{i\hbar} \right) \langle n_r(\tau_1) \rangle \left\langle T_\tau c_j(\tau) c_k^\dagger(\tau') \right\rangle \right\}$$

back to full Hamiltonian $\tilde{H} = H_0 + H_{int} + H_{ext}$

$$\langle \dots \rangle_{\tilde{H}} = \frac{\text{Tr}(\rho \dots)}{\text{Tr}(\rho)} \leftarrow \text{includes a denominator}$$

$$= \frac{1}{i\hbar} \left\langle T_\tau n_r(\tau_1) c_j(\tau) c_k^\dagger(\tau') \right\rangle_{\tilde{H}} \frac{ie}{\hbar} + \underbrace{\langle n_r(\tau_1) \rangle}_{-\langle q_r(\tau_1) \rangle} G_{jk}^\phi(\tau, \tau') \left(\frac{1}{i\hbar} \right)$$

In the equation of motion for G^ϕ , we need the term $\frac{1}{i\hbar} \left\langle T_\tau \sum_r n_r(\tau) c_j(\tau) c_k^\dagger(\tau') \right\rangle$ on page 279.

As stated earlier, as far as counter order is concerned the product of the operators $\underbrace{n c}$ must behave as a single identity.

This means we always need to put n to the left of c . This can be achieved if the time for n is slightly late than c , so we take $\tau_1 = \tau^+$

The equation of motion becomes

$$i\hbar \frac{\partial}{\partial \tau} G_{jk}(\tau, \tau') = \delta(\tau, \tau') \delta_{jk} + \sum_s H_{js}^\phi G_{sk}(\tau, \tau') + e^2 \sum_l V_{jl} \left[\frac{\delta G_{jk}(\tau, \tau')}{\delta \phi_l(\tau^+)} + \underbrace{\langle \tilde{n}_l(\tau) \rangle}_{\text{Hartree term}} G_{jk}(\tau, \tau') \right]$$

$$\tau^+ = \tau + \Delta$$

$$\text{Note } \delta G = \sum_l \delta \tau \frac{\delta G}{\delta \phi_l} \delta \phi_l \quad \text{so } \left[\frac{\delta G}{\delta \phi} \right] = \left[\frac{G}{\phi} \right]$$

$e^2 \sum_l V_{jl} \langle \tilde{n}_l \rangle$ is Hartree term

$$\text{so } i\hbar \frac{\partial}{\partial \tau} G_{jk}(\tau, \tau') = \left(\sum_s H_{js}^\phi G_{sk}(\tau, \tau') + e^2 \left(\sum_l V_{jl} \langle \tilde{n}_l \rangle \right) G_{jk}(\tau, \tau') + \frac{e\hbar}{i} \sum_l V_{jl} \frac{\delta}{\delta \phi_l(\tau^+)} G_{jk}(\tau, \tau') \right) = \delta(\tau, \tau') \delta_{jk}$$

We can write this in matrix/vector form as

$$\left[i\hbar \frac{\partial}{\partial \tau} \mathbf{I} - \mathbf{H}^{\text{tot}} + i e \hbar \left\{ \mathbf{V} \cdot \frac{\delta}{\delta \phi(\tau^+)} \right\} \right] \mathbf{G}(\tau, \tau') = \delta(\tau, \tau') \mathbf{I}$$

$$\mathbf{H}_{jk}^{\text{tot}} = H_{jk} + \delta_{jk} \left[(-e) \phi_j + e^2 \sum_l V_{jl} \langle \tilde{n}_l \rangle \right]$$

$\underbrace{\hspace{10em}}_{(-e)V(q)}$

$$= H_{jk} + \delta_{jk} V_j^{\text{tot}} (-e)$$

$$V_j^{\text{tot}} = \phi_j + \underbrace{\left(V \langle q \rangle \right)}_{\text{matrix}} \leftarrow \text{Hartree potential}$$

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By taking the functional derivative of G , we were able to convert the nonlinear term $\langle T_\tau (n(\tau) c(\tau) c^\dagger(\tau')) \rangle$ into a linear form and the equation

$$\left(i\hbar I \frac{\partial}{\partial \tau} - \tilde{H} \right) G(\tau, \tau') = I \delta(\tau, \tau')$$

here $\tilde{H}$ is a matrix and differential operator on $\mathcal{Q}$

$$\tilde{H} = H + \text{diagonal term} \quad \text{external potential}$$

$$\text{the } j^{\text{th}} \text{ diagonal term} = (-e) \left[\mathcal{Q}_j + [V \langle q \rangle]_j \right] \quad \text{hartree term}$$

$$+ i\hbar \left[V \cdot \frac{\delta}{\delta \mathcal{Q}} \right]_j$$

functional derivative

$$\left[V \cdot \frac{\delta}{\delta \mathcal{Q}} \right]_j = \sum_l V_{jl} \frac{\delta}{\delta \mathcal{Q}_l}$$

This is a very important equation, since, it makes G "closed". But the price we pay is that it is in a very high-dimensional space, since G should be viewed as the complete function of $\mathcal{Q}_j(\tau)$.

This equation, according to Hedin, is originally due to Schwinger, 1959. Here the equation is expressed in slightly different notation, is equivalent to (A10) in Hedin's Phys. Rev. paper (1965).

The next few steps is to identify the self energy Σ and get rid of $\mathcal{Q}$ in favour of other variable. Compare the differential form of the Dyson equation, we are almost there except in Dyson equation, we have $-\Sigma G$, but here

$$(ie) i\hbar \left\{ V \cdot \frac{\delta}{\delta \phi} \right\}_j G(\tau, \tau') = - \Sigma G$$

full form with index

$$\rightarrow e i\hbar \sum_l V_{jl} \frac{\delta}{\delta \phi(\tau^+)} G_{jk}(\tau, \tau') = - \sum_s \sum_{js} (\tau, \tau'') G_{sk}(\tau'', \tau') d\tau''$$

The left side is $\frac{\delta G}{\delta \phi}$ but right side is $\Sigma \cdot G$

In order to have something times G , we use a trick.

to express $\frac{\delta G}{\delta \phi}$ by $\frac{\delta G^{-1}}{\delta \phi}$. Here G^{-1} means

$$\int G(\tau, \tau'') G(\tau'', \tau')^{-1} d\tau'' = I \delta(\tau, \tau')$$

multiply in matrix elements
convolute in time τ''

compare $d(\frac{1}{x}) = -\frac{1}{x^2} dx$

Symbolically,

$$G G^{-1} = 1$$

$$\text{so } \delta G G^{-1} + G \delta G^{-1} = 0$$

(Leibnitz rule)

$$\text{or } \delta G = - G \delta G^{-1} G$$

multiply G from left

$$\text{so } \frac{\delta G(\tau, \tau')}{\delta \phi(\tau_1)} = - \int G(\tau, \tau_2) \frac{\delta G^{-1}(\tau_2, \tau_3)}{\delta \phi(\tau_1)} G(\tau_3, \tau') d\tau_2 d\tau_3$$

here on the left matrix multiplication on space index in G is implied. Put this back into the left side we get

$$e i\hbar \left\{ V \cdot \frac{\delta}{\delta \phi(\tau^+)} \right\}_j G(\tau, \tau') = ?$$

To simplify notation, we use

$$1 \equiv (j, \tau_1) \quad G_{jk}(\tau_1, \tau_2) \rightarrow G(1, 2)$$

$$\text{also } V(1, 2) \rightarrow V_{j_1 j_2} \delta(\tau_1, \tau_2)$$

introduce

$$\int d3 \rightarrow \sum_{j3} \int d\tau_3$$

$$\text{so } e i\hbar \int V(1, 3) \frac{\delta}{\delta \phi(3)} G(1, 2) d3 = - \int \Sigma(1, 3) G(3, 2) d3$$

transfer τ_1 into 1.

With these short-hand notation, we have

$$e i \hbar \int d3 \left[v(1,3) \left[-G(1,4) \frac{\delta G^{-1}(4,5)}{\delta \varphi(3)} G(5,2) \right] d4 d5 \right. \\ \left. = - \int d3 \Sigma(1,3) G(3,2) \right]$$

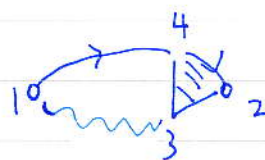
Rename 5 as 3, $3 \rightarrow 4$
we identity

$$\Sigma(1,3) = e i \hbar \int \int v(1,5) G(1,4) \frac{\delta G^{-1}(4,3)}{\delta \varphi(5)} d5 d4$$

or remove the number again $3 \rightarrow 2$, $5 \rightarrow 3$

$$\Sigma(1,2) = i \hbar e \int d3 \int d4 v(1,3) G(1,4) \frac{\delta G^{-1}(4,2)}{\delta \varphi(3)}$$

This is a formal exact expression for electron self energy. Equivalent to Eq (A16)



We should think of $e = -1$ in Hedin's notation.

Our next step is to evoke the theory of screening. So the interaction v becomes the screened one, we called D , the full photon Green's function. Satisfies the Dyson equation, symbolically

$$D = v + v \Pi D \quad D \rightarrow D_{jk}(\tau, \tau')$$

here we take this

$$v \rightarrow V_{jk} \delta(\tau, \tau')$$

as a definition and try to define an equation for the self energy Π . We define the dielectric constant (matrix as)

$$\epsilon = 1 - v \Pi$$

$$\text{so } \epsilon D = v$$

$$\text{and define } \epsilon^{-1} = 1 + v \chi$$

Here χ is _{charge} density correlation function

The relation between χ and Π is

$$\chi = \Pi \epsilon^{-1} = \Pi + \Pi \chi$$

ϵ has been defined such that

$$\text{and } \epsilon^T q = q^{\text{ex}} \leftarrow \text{external charge}$$

↑
induced charges
actual

$$\begin{array}{ccc} \epsilon \phi = \phi^{\text{ex}} & & \\ \text{actual} & \text{external} & \\ \downarrow & \downarrow & \\ V^{\text{tot}} & \phi & \text{new notation} \end{array}$$

For further see Mahan-1-Yu. poly.

Here $\chi_{jk}(\tau, \tau') = -\frac{i}{\hbar} \langle T_\tau \delta q_j(\tau) \cdot \delta q_k(\tau') \rangle$

here $\delta q = q - \langle q \rangle$ we need a proof that this is true. χ and Π is symmetric, like D , but ϵ is not.

The total actual potential is

$$V_j^{\text{tot}}(\tau) = \phi_j(\tau) + \sum_k V_{jk} \langle q_k \rangle(\tau)$$

or symbolically $V_{\text{tot}}(1) = \phi(1) + \int d2 \, v(1,2) \langle \hat{q}(2) \rangle$

Since $\epsilon \phi = \phi^{\text{ex}} \Rightarrow \int \epsilon(1,2) \delta V_{\text{tot}}(2) d2 = \delta \phi(1)$

$$\frac{\delta \phi(1)}{\delta V_{\text{tot}}(2)} = \epsilon(1,2)$$

$$\epsilon^{-1}(1,2) = \frac{\delta V_{\text{tot}}(1)}{\delta \phi(2)}$$

I hope the notation is consistent

$$\begin{aligned} \text{since } \int \epsilon^{-1}(1,2) \epsilon(2,3) d2 &= \int d2 \left[\frac{\delta V_{\text{tot}}(1)}{\delta \phi(2)} \frac{\delta \phi(2)}{\delta V_{\text{tot}}(3)} \right] \\ &= \delta(1,3) \end{aligned}$$

With this relation (or definition of) ϵ^{-1} to $\frac{\delta V_{\text{tot}}}{\delta \phi}$, we get

$$\epsilon^{-1}(1,2) = \frac{\delta V_{\text{tot}}(1)}{\delta \phi(2)} = \delta(1,2) + \int d3 \, v(1,3) \frac{\delta \langle q(3) \rangle}{\delta \phi(2)}$$

We keep ϕ small, to 1 (linear order in ϕ), we can apply the linear response theory so

$$\frac{\delta \langle q(3) \rangle}{\delta \varphi(2)} = \chi(3, 2)$$

since

$$\delta \langle q(3) \rangle = \int \chi(3, 2) \delta \varphi(2) d2$$

thus $\epsilon^{-1} = 1 + v\chi$ is verified and χ is indeed the fluctuation of charge.

The standard notation for the screened potential is W which I called D . Let us W . Then

$$W = \epsilon^{-1} v = (1 + v\chi)v = v + v\chi v$$

$$= v + \Pi W$$

We need to change v to W in the self energy Σ formula. To do this we make a change of variable from φ to $V^{\text{tot}} = \varphi + \int d2 v(1, 2) \langle \hat{q}_2 \rangle$

so $\rightarrow$ use the chain rule: $\frac{\delta}{\delta \varphi(3)} = \left(\frac{\delta V^{\text{tot}}(3)}{\delta \varphi(3)} \frac{\delta}{\delta V^{\text{tot}}(3)} \right) d4$

$$= \int d4 \epsilon^{-1}(4, 3) \frac{\delta}{\delta V^{\text{tot}}(4)}$$

we have:

$$\rightarrow W(1, 2) = \int d3 \epsilon^{-1}(1, 3) v(3, 2)$$

We define the vertex function as $\Gamma(1, 2; 3) = - \frac{\delta G^{-1}(12)}{\delta V^{\text{tot}}(3)}$

using these relations we have

using the fact that v , Π , χ and thus are symmetric with respect to the two arguments 1 and 2 we can also write

~~$W = v \epsilon^{-1}$~~ wrong

~~$W(1, 2) = \int d3 v(1, 3) \epsilon(3, 2)$~~

Using the above relation, we are ready to write down the form of the Hedin equations for the electron self energy $\Sigma(1, 2)$ as

$$\Sigma(1,2) = i\hbar e \int d^3 \int d^4 v(1^+, 3) G(1, 4) \frac{\delta G(4, 2)^{-1}}{\delta \varphi(3)}$$

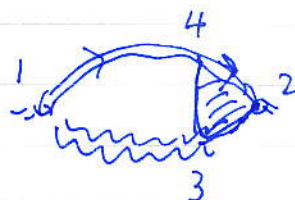
$$= (-e) i\hbar \int d^3 \int d^4 \int d^5 \underline{v(1^+, 3)} G(1, 4) \underline{E^{-1}(5, 3)} \Gamma(4, 2; 5)$$

$$= (-e) i\hbar \int d^4 \int d^5 W(1^+, 5) G(1, 4) \Gamma(4, 2; 5)$$

rename $5 \rightarrow 3$

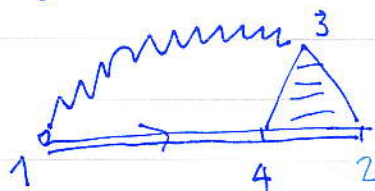
$[v(3, 1^+) = v(1^+, 3), W \text{ is symmetric}]$
 $E \text{ is not symmetric}$

$$\Sigma(1, 2) = (-e) i\hbar \int d^3 \int d^4 W(1^+, 3) G(1, 4) \Gamma(4, 2; 3)$$



This result is known before Hedin. eg. in AGD, by purely diagrammatic argument. Hedin obtained the same result analytically/algebraically. See page 88, 89 in AGD [Abrikosov, Gorkov, Dzyaloshinski]. This result was also presented diagrammatically in Fetter and Walecka.

Both diagram



To determine W , we need Π , by

$$W = v + v \Pi \cdot W$$

We thus need an exact analytic expression for the polarization diagram $\Pi(1, 2)$. We can get a functional derivative for Π as

$$W = v E^{-1} \Rightarrow W(1, 2) = v(1, 3) \frac{\delta V_{\text{int}}(3)}{\delta \varphi(2)} d^3$$

$$= v(1, 3) \left[\delta(3, 2) + \int d^4 v(3, 4) \frac{\delta \langle \hat{q}(4) \rangle}{\delta \varphi(2)} \right] d^3$$

$$= v(1, 2) + \int d^3 d^4 d^5 v(1, 3) v(3, 4) \frac{\delta \langle \hat{q}(4) \rangle}{\delta V_{\text{int}}(5)} \frac{\delta V_{\text{int}}(5)}{\delta \varphi(2)}$$

$$= V(1,2) + \int d^3 d^4 \cancel{\delta(1,3)} \frac{\delta \langle \hat{q}(4) \rangle}{\delta V_{tot}(5)}$$

$$W = E^{-1} V$$

$$W(1,2) = \int E^{-1}(1,3) V(3,2) d^3 = \int \frac{\delta V_{tot}(1)}{\delta \varphi(3)} V(3,2) d^3$$

$$= \int d^3 \left[\delta(1,3) + \int d^4 V(1,4) \frac{\delta \langle q(4) \rangle}{\delta \varphi(3)} \right] V(3,2)$$

$$= V(1,2) + \int d^4 V(1,4) \frac{\delta \langle q(4) \rangle}{\delta \varphi(3)} V(3,2)$$

Compare with $W = V + V \chi V$ we see $\chi(1,2) = \frac{\delta \langle q(1) \rangle}{\delta \varphi(2)}$

$$= V(1,2) + \int d^4 d^5 V(1,4) \frac{\delta \langle q(4) \rangle}{\delta V_{tot}(5)} \frac{\delta V_{tot}(5)}{\delta \varphi(3)} V(3,2) \quad \text{as before}$$

$$= V(1,2) + \int d^4 \int d^5 V(1,4) \frac{\delta \langle q(4) \rangle}{\delta V_{tot}(5)} \underbrace{E^{-1}(5,3)}_{\parallel W(5,2)} V(3,2)$$

$5 \rightarrow 4 \quad 4 \rightarrow 3$

$$= V(1,2) + \int d^3 \int d^4 V(1,3) \frac{\delta \langle q(3) \rangle}{\delta V_{tot}(4)} W(4,2) \quad E = 1 - V \Pi$$

so $\Pi(3,4) = \frac{\delta \langle q(3) \rangle}{\delta V_{tot}(4)}$ so Π maps total potential to induced charge

See Mahan-L-Yu. p. 44 page 193 for this interpretation.

To calculate this functional derivative

$$\langle q(1) \rangle = \langle e \rangle (-i\hbar) G(1,1^+)$$

$$G^< = \frac{i}{\hbar} \langle c^\dagger c \rangle$$

$$\text{but } \frac{\delta G(1,2)}{\delta V_{tot}(3)} = - \int d^4 d^5 G(1,4) \frac{\delta G^<(4,5)}{\delta V_{tot}(3)} G(5,2)$$

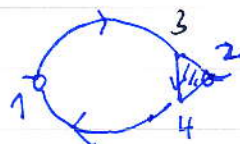
again this is because $GG^{-1} = I$ and $\delta G G^{-1} + G \delta G^{-1} = 0$

$$\text{so } \Pi(1,2) = \frac{\delta \langle q(1) \rangle}{\delta V_{tot}(2)} = e i \hbar \frac{\delta G(1,1^+)}{\delta V_{tot}(2)} = - e i \hbar \int d^4 d^5 \left[G(1,4) \frac{\delta G^<(4,5)}{\delta V_{tot}(2)} G(5,1^+) \right]$$

using the definition for Γ , $\Gamma(1,2;3) = - \frac{\delta G^{-1}(1,2)}{\delta V_m(3)}$

we get $\Pi(1,2) = e i \hbar \int d^3 d^4 G(1,3) \Gamma(3,4;2) G(4,1^+)$

This is the 2nd Heisenberg equation for the self energy of photon.



We see that we started with ϕ as the fundamental variable but the transformation to V_{tot} is an important step to take.

The last equation is a self-consistent equation for Γ itself.

The logic is like this. Originally we have one functional derivative equation for the Green's function G^{ϕ} , as an equation of motion as given on page 287, 289. However, this functional derivative equation is in a very high dimension (G is a function of each $\phi(x)$).

Can we solve such equation on computer?

With the help of new variable $V_{tot}(1) = \phi(1) + \int v(1,2) \langle q(2) \rangle d^2$ and the dielectric function

$\epsilon^{-1}(1,2) = \frac{\delta V_{tot}(1)}{\delta \phi(2)}$, we can express two self-energies

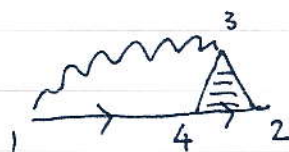
the electronic one and photon one by the vertex function

$$\Gamma(1,2;3) = - \frac{\delta G^{-1}(1,2)}{\delta V_{tot}(3)}$$

Diagrammally,

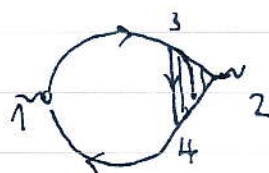
this is

$$\Sigma(1,2) = (-e) i \hbar$$



$$G(1,2) = \text{fermion line from 1 to 2}$$

$$\Pi(1,2) = e i \hbar$$



$$W(1,2) = \text{photon line from 1 to 2} \\ = W(2,1)$$

$$\Gamma(1,2;3) = \text{shaded triangle with vertices 1, 2, 3}$$

However, the complexity does not go away. It is now reside in Γ . To get an equation for the vertex function $\Gamma(1,2;3)$ we need the Dyson equation

$$G = G_0 + G_0 (\{V_{tot}\}(-e)\hbar \Sigma) G$$

Here G_0 is from $c^\dagger H c$ not including the external field g and Hartree term vq . We have put these two terms in $V_{tot} = g + vq$. This term acting only as a diagonal matrix elements in space j and time τ so the curly braces $\{V_{tot}\}$. From the above we find the inverse as

$$G^{-1}(1,2) = G_0^{-1}(1,2) + e V_{tot}(1) \delta(1,2) - \Sigma(1,2)$$

$\delta(1,2) \rightarrow \delta_{j,k} \delta(\tau_1, \tau_2)$ is the δ -function in space & time. Take variation, since G_0^{-1} does not depend on g , and $V_{tot}(1)$ is to be the independent variable for the variation we get

$$\Gamma(1,2;3) = - \frac{\delta G^{-1}(1,2)}{\delta V_{tot}(3)} = (-e) \delta(1,3) \delta(1,2) + \frac{\delta \Sigma(1,2)}{\delta V_{tot}(3)}$$

If we omit the 2nd term, with $\Gamma(1,2;3) = -e \delta(1,2) \delta(1,3)$ we simply recover the self-consistent Born approximation (SCBA). But there is a slight difference. The Hartree term is still $g + \int v \langle q \rangle$ (with $g=0$).

not $\int W \langle q \rangle$! I.e. the potential should not be renormalized or screened, i.e. D should be $D_0 = V$. As we see the Dyson equation is the form above is an exact equation. There is no reason to change v in V_{tot} to $W = D$!

To transfer the functional derivative from $\frac{\delta \Sigma}{\delta V_{tot}}$ to $\frac{\delta \Sigma}{\delta G}$ we need to think what does Σ depend on.

If we do the standard diagrammatic expansion the self-energy is expressed by the non-interacting Green's functions G_0^g and V . V does not depend on g . g is transformed to a new variable V_{tot} one-to-one. However, we need the full Green's function G^g . We assume we can invert the Dyson equation

$$G = G_0 + G_0 \Sigma(G_0, V) G$$

taking V as parameter i.e. $G_0 \equiv G_0(G, V)$

Substitute this back to the self energy $\Sigma(G_0, V)$ we can express it as solely a function of G and V . i.e.

$$\Sigma = \Sigma(G_0, V) = \Sigma(G, V)$$

since V is independent of g thus V_{tot} we have the chain rule

$$\frac{\delta \Sigma(1, 2)}{\delta V_{\text{tot}}(3)} = \int d4 d5 \frac{\delta G(4, 5)}{\delta V_{\text{tot}}(3)} \frac{\delta \Sigma(1, 2)}{\delta G(4, 5)}$$

However, if we express the self energy as $\Sigma(G, W)$ as in Hedén's original paper, then W also depends on g the chain rule is not so simple as we also need $\frac{\delta W}{\delta V_{\text{tot}}}$. Then

$$\frac{\delta \Sigma(1, 2)}{\delta V_{\text{tot}}(3)} = \int d4 \int d5 \left[-G(4, 6) \frac{\delta G(6, 7)}{\delta V_{\text{tot}}(3)} G(7, 5) \frac{\delta \Sigma(1, 2)}{\delta G(4, 5)} \right] d(4, 5, 6, 7)$$

$$= \int d(4, 5, 6, 7) G(4, 6) \Gamma(6, 7; 3) G(7, 5) \frac{\delta \Sigma(1, 2)}{\delta G(4, 5)}$$

so $\Gamma(1, 2; 3) = (-e) \delta(1, 3) \delta(1, 2) + \text{above term}$

This is the last Hedén equation

Diagrammatically we have for the vertex function

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$$\begin{array}{c} \text{Diagram 1: Triangle with 3 external lines (1, 2, 3)} \end{array} = (-e) \begin{array}{c} \text{Diagram 2: Triangle with 3 external lines (1, 2, 3) and a cross inside} \end{array} + \begin{array}{c} \text{Diagram 3: Triangle with 3 external lines (1, 2, 3) and a cross inside} \end{array}$$

two δ functions

square = $\frac{\delta \Sigma(1,2)}{\delta G(4,5)}$

Starting page 263, we defined the tight-binding model with Coulomb interaction (with self-interaction removed). We then derive the equation of motion for G , replacing the nonlinear term by functionally derivatives. This is a self-contained equation. However, we prefer an expression for self energy. A trick is used to transfer δG to δG^{-1} by noting $GG^{-1} = I$. Then we need the physics of screening, i.e. $\epsilon = 1 - v\Pi$, $\frac{1}{\epsilon} = 1 + v\chi$, etc. From this we can change from g to V_{tot} and expresses two self-energies in terms of G , W and vertex function F . Finally, we get an equation for Π . I think I have derived Hedin equations with sufficient details and justifications.

We have demonstrated a detailed derivation of the Hedin equations for the free electron + Coulomb interaction system. Our next task is to give a derivation of the electron-phonon system (without electron-electron interaction).

Since electron-phonon interaction is structurally identical to electron-scalar field system, I also expect identical result for the phonon system with D takes the position of W .

However, there is one puzzling aspect in the phonon case. There is no equivalence of the potential $V_{tot}(r)$ and hartree term in phonon case. We need to see how these differences are resolved.

We consider a finite cluster with a tight-binding electron system $\hat{H}_e = c^\dagger H c$ here H is a

finite N by N hermitian matrix $H^\dagger = H \rightarrow H_{ik}^* = H_{kj}$

$$c = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix} \quad c^\dagger = (c_1^\dagger, c_2^\dagger, \dots, c_N^\dagger) \text{ is row vector.}$$

The free phonon is $\hat{H}_p = \frac{1}{2} P^T P + \frac{1}{2} U^T K U$ here K is real and symmetric $K^T = K$, we call the matrix K Dynamic matrix and $U = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_j \end{pmatrix} \equiv \vec{u}$ is a finite column vector. And the equation of motion for the phonon takes the usual Hamilton form

$$\dot{u} = \frac{\partial H_p}{\partial p} \quad \dot{p} = - \frac{\partial H_p}{\partial u}$$

$$\text{or} \quad \dot{u} = P, \quad \dot{p} = \ddot{u} = -K u$$

Electron-phonon interaction is $H_{ep} = \sum M_{jk}^2 c_j^\dagger c_k u_e$
 $= c^\dagger \vec{M} c \cdot \vec{u}$

We do not assume periodicity. The advantage of such a notation is its generality. If the system does have periodicity, we can incorporate such fact at the very end, which only means the eigen modes, defined by

$$S^\dagger H S = \begin{pmatrix} \epsilon_1 & & 0 \\ & \ddots & \\ 0 & & \epsilon_i \end{pmatrix} \quad T^T K T = \begin{pmatrix} \omega_1^2 & & 0 \\ & \ddots & \\ 0 & & \omega_n^2 \end{pmatrix}$$

is such that the wave vector $\vec{k}$ is a good "quantum number".
 Here $S^\dagger S = S S^\dagger = I, \quad T^T T = T T^T = I$

Our task is to obtain the associated Hedin equations for the Hamiltonian $\hat{H} = c^\dagger H c + \frac{1}{2}(p^2 + u^\dagger K u) + c^\dagger \vec{M} c \cdot \vec{u}$

which relate the Green's functions G , D , and self-energies Σ and Π , and finally the vertex function Γ . I claim it is nearly the same as before for the electron-Coulomb case but we need a proof.

Here we sort of follow the notes by Yu Ya dated 14 May 2017 (<http://beside.web.fc2.com/>), entitled "Hedin equations for interacting Electron-phonon systems".

In order to dynamically perturb the system, we introduce a contour dependent force $f_e(\tau)$ coupled to the phonon coordinates u_e , i.e. We add one extra term to the Hamiltonian $\hat{H}$

$$\hat{H}' = - \sum_e f_e(\tau) \hat{u}_e$$

$\hat{u}_e$ is in Schrödinger picture and in $f(\tau)$, τ is explicit time (time dependent force). Since $[\hat{u}] = \sqrt{M} \cdot L$, the units for f is $[f] = \frac{E}{\sqrt{M}}$ so that $[f u]$ is energy.

Here $f_e(\tau)$ plays the role of q (see page 265) coupled to the charge $\hat{q}$ in the Coulomb case. The role of V_{tot} is now played by $\langle \hat{u}_e \rangle$ that is conjugate to f (very much like the $\langle q \rangle$).

Our first step is to derive the equation of motion for the Hamiltonian $\hat{H} + \hat{H}' = c^\dagger H c + \frac{1}{2}(p^2 + u^\dagger K u) + c^\dagger \vec{M} c \cdot \vec{u} - \vec{f}(\tau) \cdot \vec{u}$

Since $i\hbar \frac{d}{dt} c = [c, \hat{H} + \hat{H}'] = H c + \vec{M} c \cdot \vec{u}$ the extra perturbation does not give new terms.

while $\frac{d^2 u}{dt^2} = \frac{dp}{dt} = \frac{1}{i\hbar} [p, \hat{H} + \hat{H}'] = -K u - c^\dagger M c - f$

all three terms are vectors

From these, we obtain the equation of motion for the counter ordered Green's function as

$$i\hbar \frac{\partial}{\partial \tau} G_{jk}(\tau, \tau') - \sum_s \hat{H}_{js} G_{sk}(\tau, \tau') - (-\frac{i}{\hbar}) \sum_{ls} M_{ls}^j \langle T_\tau c_l(\tau) c_s^\dagger(\tau) u_k(\tau) \rangle = \delta_{jk} \delta(\tau, \tau')$$

$$\frac{\partial^2}{\partial \tau^2} D_{jk}(\tau, \tau') + \sum_s \hat{K}_{js} D_{sk}(\tau, \tau') + (-\frac{i}{\hbar}) \sum_{ll'} M_{ll'}^j \langle T_\tau (c_l^\dagger(\tau) c_{l'}(\tau)) u_k(\tau) \rangle = -\delta_{jk} \delta(\tau, \tau')$$

$$+ (-\frac{i}{\hbar}) f_j(\tau) \langle u_k(\tau) \rangle$$

Here the Green's function are defined in the usual way

$$G_{jk}(\tau, \tau') = -\frac{i}{\hbar} \langle T_\tau c_j(\tau) c_k^\dagger(\tau') \rangle$$

$$D_{jk}(\tau, \tau') = -\frac{i}{\hbar} \langle T_\tau (u_j + \langle u_j \rangle) (u_k(\tau') - \langle u_k(\tau') \rangle) \rangle$$

Since $\langle u_j(\tau) \rangle \neq 0$, we subtract off so that D is a Dyson equation

$$\langle \dots \rangle = \frac{\text{Tr}(\rho \dots S)}{\text{Tr}(\rho S)} \quad S = T_\tau e^{-\frac{i}{\hbar} \int (-f_j(\tau') u_j(\tau')) d\tau'}$$

here S is governed by $\hat{H}'$ only and the operators $c, c^\dagger, u$ evolving according $\hat{H}$. We see

$$\delta S = T_\tau \sum_j \int d\tau'' \left(\frac{i}{\hbar} \delta f_j(\tau'') (u_j(\tau'')) \right) \cdot S$$

or $\frac{\delta S}{\delta f_j(t)} = T_\tau \left(\frac{i}{\hbar} u_j(t) S \right)$ This analogy to Eq. $\frac{\delta S}{\delta g}$ on page 285.

We now ask how does the average position $\langle u_j(\tau) \rangle$ changes with the application of a time dependent force $f_k(\tau')$? The answer is

precisely given by linear response theory with the response function $D(\mathbf{r}, \mathbf{r}')$. We derive this result by the definition of the average $\langle u \rangle$ and use functional derivative. We note that $\langle u \rangle$ is not constant since f is time dependent.

To work out this linear response since $\langle u \rangle_{\hat{H}} \neq 0$ in general even when $f \equiv 0$ but $\langle u \rangle$ under the total Hamiltonian is a constant, we consider

$$\langle \Delta u \rangle = \langle u \rangle_{\hat{H} + \hat{H}'} - \langle u \rangle_{\hat{H}} = \langle (u - \langle u \rangle_{\hat{H}}) \rangle_{\hat{H} + \hat{H}'}$$

finite f $f \equiv 0$ for all $\mathbf{r}$ and j

So $\Delta u = u - \langle u \rangle_{\hat{H}}$ the fluctuation is with respect to the time independent full Hamiltonian $\hat{H} = \hat{H}_e + \hat{H}_p + \hat{H}_{ep}$.

so

$$\langle \Delta u(t) \rangle = \frac{\text{Tr}(\rho T_{\tau}(u(t) - \langle u \rangle_{\hat{H}}) S')}{\text{Tr}(\rho S')}$$

Now we make variation with respect to f so

$$\delta \langle \Delta u(t) \rangle = \frac{\text{Tr}(\rho T_{\tau}(u(t) - \langle u \rangle_{\hat{H}}) \delta S')}{\text{Tr}(\rho S')} - \frac{\text{Tr}(\rho (u - \langle u \rangle) \delta S)}{\text{Tr}(\rho S)^2} \text{Tr}(\rho S S')$$

Note that in the interaction picture with interaction as $H' = - \sum_j f_j(\mathbf{r}) u_j$ and the Heisenberg evolution by $\hat{H}$ for the operator $u(\mathbf{r}) \rightarrow u(t) = e^{+\frac{i}{\hbar} \hat{H} t} u e^{-\frac{i}{\hbar} \hat{H} t}$ the only f dependence is in S and we already get

$$\delta S = \left(\frac{i}{\hbar} \right) \int d\mathbf{r} u(\mathbf{r}) \delta f(\mathbf{r}) \cdot S$$

$$j \equiv (j, \tau)$$

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$$\delta \langle \Delta U(j) \rangle = \int d^2 \frac{i}{\hbar} \delta f(z) \left[\frac{\text{Tr}(\rho T_z U(j) U(z) \cdot S)}{\text{Tr}(\rho S)} - \frac{\text{Tr}(\rho \Delta U(j) S)}{\text{Tr}(\rho S)} \right]$$

It is clear $\delta \langle U \rangle_H = 0$
 It is just a number so not variation w.r.t f

$$\text{so } \delta F \equiv F(f + \delta f) - F(f)$$

$$\begin{aligned} \delta \langle \Delta U(1) \rangle &= \delta \langle U(1) \rangle = \int d^2 \left(\frac{i}{\hbar} \right) \delta f(z) \left[\langle T_z U(1) U(z) \rangle - \langle U(1) \rangle \langle U(z) \rangle \right] \\ &= - \int d^2 D(1, z) \delta f(z) \end{aligned}$$

here $D(1, z) = \left(-\frac{i}{\hbar} \right) \langle T_z (U(1) - \langle U(1) \rangle) (U(z) - \langle U(z) \rangle) \rangle$

Note $\Delta U(1) = U(1) - \langle U(1) \rangle$ only when $f \equiv 0$. In the definition of D $\langle U(1) \rangle$ is over arbitrary f , i.e.

$$\langle \dots \rangle = \frac{\text{Tr}(\rho \dots S)}{\text{Tr}(\rho S)} \quad \text{in which } S \text{ contains } f.$$

This result recovers the usual linear response theory result if we say that the force $f(\tau) = f^\sigma(t)$ is independent of the branch index σ . Then

$$\delta \langle \Delta U^\pm(t) \rangle = - \left[\int_{-\infty}^{\infty} dt' D^+(t, t') f(t') - \int_{-\infty}^{\infty} dt' D^-(t, t') f(t') \right]$$

remember $\int d\tau' = \sum_{\sigma} \int_{-\infty}^{\infty} dt' \cdot \sigma \quad \sigma = \pm$

$$\delta \langle \Delta U^\pm(t) \rangle = - \int_{-\infty}^{\infty} dt' [D^+(t, t') - D^-(t, t')] f(t')$$

$$+ \begin{bmatrix} D^+ & D^- \\ D^+ & D^- \end{bmatrix} = - \int_{-\infty}^{\infty} dt' D^r(t, t') f(t')$$

This is precisely the retarded Green's function.

According to the definition of functional derivative we have

$$\frac{\delta \langle u(1) \rangle}{\delta f(2)} = -D(1, 2)$$

Since we have the general relation $D^+ + D^- = D^> + D^<$
or $D^> = D^+ - D^- = D^> - D^-$ we see the response of the position of atoms is independent of the branch index σ

$$\begin{aligned} \delta \langle u(t) \rangle &= \delta \langle u^+(t) \rangle \rightarrow \delta u(t) = \sum_{\sigma'} \int dt' D(t, t')^{\sigma \sigma'} f(t') \\ &= - \int_{-\infty}^{+\infty} dt' \left[\underset{\substack{\downarrow \\ D^>}}{D^+(t, t')} - D^-(t, t') \right] f(t') = - \int_{-\infty}^{+\infty} D^>(t, t')^V dt' f(t') \end{aligned}$$

We now work out the functional derivatives of the contour ordered Green's functions $\frac{\delta G}{\delta f}$, $\frac{\delta D}{\delta f}$ and see how it is related to the nonlinear terms in the equation of motion for the Green's functions. We write out the Green's function in the form

$$G(1, 2) = -\frac{i}{\hbar} \frac{\langle T_c [c(1) c^\dagger(2) S'] \rangle}{\langle S \rangle} \quad \text{here we use } \langle \dots \rangle = \text{Tr}(\rho \dots)$$

$$S' = e^{-\frac{i}{\hbar} \int (-f(3) u(3)) d3}$$

$$d3 \equiv \sum_{j_3} \int d\tau_3$$

The contour order super operator apply to all the operators to the right of it.

Here $1 \equiv (j, \tau_1)$ etc. These $c(1)$, $u(3)$ etc are in the Heisenberg picture governed by full Hamiltonian $\hat{H}$ on page 313. So the only dependence of the time-dependent force is in the scattering "matrix" S' .

We make the variation with respect to G , obtain

$$\delta G(1,2) = -\left(\frac{i}{\hbar}\right) \left[\frac{\langle T_c(c(1) c(2)^\dagger) \delta S \rangle}{\langle S \rangle} - \frac{\langle T_c(c(1) c(2)^\dagger) S \rangle \langle \delta S \rangle}{\langle S \rangle^2} \right]$$

Now $\delta S = \frac{i}{\hbar} \int d^3z f(z) U(z) d^3 \cdot S$

so $\delta G(1,2) = \int d^3z \delta f(z) \left(\frac{i}{\hbar}\right) \left[-\left(\frac{i}{\hbar}\right) \frac{\langle T_c[c(1) c(2)^\dagger] U(z) S \rangle}{\langle S \rangle} + \frac{i}{\hbar} \frac{\langle T_c(c(1) c(2)^\dagger) S \rangle}{\langle S \rangle} \frac{\langle U(z) S \rangle}{\langle S \rangle} \right]$

$\leftarrow$ this is the definition of functional derivative

$$= \int d^3z \frac{\delta G(1,2)}{\delta f(z)} \cdot \delta f(z)$$

Thus $\frac{\delta G(1,2)}{\delta f(z)} = \dots$ $\frac{1}{\hbar^2} = \frac{i}{\hbar} \cdot (-\frac{i}{\hbar})$

$$= \frac{1}{\hbar^2} \langle T_c(c(1) c(2)^\dagger) U(z) \rangle_{H+H'} + \left(-\frac{i}{\hbar}\right) G(1,2) \langle U(z) \rangle_{H+H'}$$

Here $\langle \dots \rangle_{H+H'} = \frac{\langle \dots S \rangle}{\langle S \rangle}$ is the average with respect to the total Hamiltonian $H+H'$, $H' = -\vec{f} \cdot \vec{u}$

Compare with the equation of motion we get

$$i\hbar \frac{\partial}{\partial \tau} G_{jk}(\tau, \tau') - \sum_s H_{js} G_{sk}(\tau, \tau') + \sum_s M_{js}^2 \left(\frac{\delta G_{sk}(\tau)}{\delta f(\tau)} - G_{sk}(\tau, \tau') \langle u_s(\tau) \rangle \right) = \delta_{jk} \delta(\tau, \tau')$$

Note that we got a new term $\langle u_s(\tau) \rangle$ here (Hartree term)

$\Delta u = u - \langle u \rangle$

similarly $D(1,2) = -\frac{i}{\hbar} \frac{\langle T_c \Delta u(1) \Delta u(2) S \rangle}{\langle S \rangle}$ and

$$\delta D(1,2) = -\frac{i}{\hbar} \left[\frac{\langle T_c \Delta u(1) \Delta u(2) \delta S \rangle}{\langle S \rangle} - \frac{\langle T_c \Delta u(1) \Delta u(2) S \rangle \langle \delta S \rangle}{\langle S \rangle^2} \right]$$

Since $\delta S \propto u$ we get a $\langle T_c u u u \rangle$. Such term is not needed.

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We can write the equation of motion for G in a more compact notation with matrices, it is

$$\left[i\hbar \frac{\partial}{\partial \tau} \cdot I - \left(H + \vec{M} \cdot \langle \vec{u}(\tau) \rangle - i\hbar \vec{M} \cdot \frac{\delta}{\delta \vec{f}(\tau)} \right) \right] G(\tau, \tau') = I \delta(\tau, \tau')$$

Here $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \\ & & \ddots \end{pmatrix}$ is $N \times N$ identity matrix

H is the single particles hamilton. The usual hartree term (see page 273) of $v(q)$ is now $\vec{M} \cdot \langle \vec{u} \rangle$, which is not m-site, but still $N \times N$ matrix by making $\vec{M} \equiv (M^l)$ dotted into the vector $\frac{\delta}{\delta \vec{f}} \equiv \left(\frac{\delta}{\delta f_l} \right)$ the new matrix operator is acting on the matrix G . The result is $\delta(1, 2) \equiv \sum_{j_1, j_2} \delta(\tau_1, \tau_2)$, the δ -function is space and contour time.

Our next task is to transform so that we can identify the self energy $\rightarrow \begin{cases} G = G_0 + G_0 \Sigma G \\ D = D_0 + D_0 \Pi D \end{cases}$

The equation of motion for D also requires the same functional derivative, not a different one. From page 315 we get

$$\frac{\partial^2}{\partial \tau^2} D_{jk}(\tau, \tau') + \sum_s K_{js} D_{sk}(\tau, \tau') + \sum_{l, l'} M_{ll'}^j \left(i\hbar \frac{\delta G_{ll'}(\tau, \tau')}{\delta f_k(\tau')} - G_{ll'}(\tau, \tau') \langle u(\tau') \rangle \right) = -\delta_{jk} \delta(\tau, \tau')$$

~~$-\left(\frac{i}{\hbar}\right) \dots f(\tau) \langle u(\tau') \rangle$~~

this two terms are cancelled due to $\Delta u = u - \langle u \rangle$!!

Note since in the interaction we have $c^\dagger M c$ we must put the 2nd τ to slight later than τ , we use $\tau^+ = \tau + \Delta\tau$.
Note we do "inner product" with G and "outer product" with u .
trace!

We can also write this equation compactly as matrix equation as

This should not be here, because in taken the derivative I fix the

$$\left[\frac{\partial^2}{\partial \tau^2} \mathbf{I} + \mathbf{K} \right] \mathbf{D}(\tau, \tau') - \left[\text{Tr}(\vec{M} \mathbf{G}(\tau, \tau')) + \frac{i}{\hbar} \vec{f}(\tau) \right] \otimes \langle \mathbf{u}(\tau') \rangle + i\hbar \text{Tr}(\vec{M} \otimes \frac{\delta}{\delta \mathbf{f}(\tau')} \mathbf{G}(\tau, \tau')) = -\mathbf{I} \delta(\tau, \tau')$$

substitut term *inner product*

(Here the trace is over the electron degrees of freedom only. The meaning of this equation is obscure, particularly the $\langle \mathbf{u}(\tau') \rangle$ term.) The two equations, $\mathbf{G}$ and $\mathbf{D}$ do not close themselves, because of the extra $\langle \mathbf{u}(\tau) \rangle$.

We thus need an equation for $\mathbf{u}$, which is

$$\frac{d^2}{d\tau^2} \langle \mathbf{u}(\tau) \rangle = -\mathbf{K} \langle \mathbf{u}(\tau) \rangle + i\hbar \text{Tr}(\vec{M} \mathbf{G}(\tau, \tau')) - \vec{f}(\tau)$$

Is this equation really necessary? Yes. In steady state $\langle \mathbf{u}(\tau) \rangle$ becomes independent of τ with $\vec{f} \equiv 0$

we need to define $\langle \vec{u} \rangle$ by

$$-\mathbf{K} \langle \vec{u} \rangle + i\hbar \text{Tr}(\vec{M} \mathbf{G}(\tau, \tau')) = 0$$

or $\langle \vec{u} \rangle = i\hbar \mathbf{K}^{-1} \text{Tr}(\vec{M} \mathbf{G}(\tau, \tau'))$

This gives the Hartree diagram 

We need a more careful derivation of equation of motion for $\mathbf{D}$

$$\Delta \mathbf{u} = \mathbf{u} - \langle \mathbf{u} \rangle$$

$$\begin{aligned} \frac{\partial}{\partial \tau} \mathbf{D}(\tau, \tau') &= -\frac{i}{\hbar} \langle \mathcal{T}_\tau \Delta \mathbf{u}_j(\tau) \Delta \mathbf{u}_k(\tau') \rangle \\ &= \frac{1}{i\hbar} \left[\theta(\tau, \tau') \langle \Delta \mathbf{u}_j(\tau) \Delta \mathbf{u}_k(\tau') \rangle + \theta(\tau', \tau) \langle \Delta \mathbf{u}_k(\tau') \Delta \mathbf{u}_j(\tau) \rangle \right] \\ \frac{\partial}{\partial \tau} \mathbf{D}(\tau, \tau') &= \frac{1}{i\hbar} \left[\delta(\tau, \tau') \langle \Delta \mathbf{u}_j(\tau) \Delta \mathbf{u}_k(\tau) - \Delta \mathbf{u}_k(\tau) \Delta \mathbf{u}_j(\tau) \rangle \right. \\ &\quad \left. + \frac{1}{i\hbar} [\theta(\tau, \tau') \langle \Delta \dot{\mathbf{u}}_j(\tau) \Delta \mathbf{u}_k(\tau') \rangle + \theta(\tau', \tau) \langle \Delta \mathbf{u}_k(\tau') \Delta \dot{\mathbf{u}}_j(\tau) \rangle] \right] \\ &= \frac{1}{i\hbar} [\delta(\tau, \tau') \cdot 0] + \frac{1}{i\hbar} \langle \mathcal{T}_\tau \Delta \dot{\mathbf{u}}_j(\tau) \Delta \mathbf{u}_k(\tau') \rangle \end{aligned}$$

So the fine derivative is

$$\frac{\partial}{\partial \tau} D_{jk}(\tau, \tau') = \frac{1}{i\hbar} \langle T_{\tau} \Delta \dot{u}_j(\tau) \Delta u_k(\tau') \rangle$$

because of the δ -
function we can set $\tau' = \tau$

Take $\frac{\partial}{\partial \tau}$ again $\frac{\partial^2}{\partial \tau^2} D_{jk}(\tau, \tau') = \delta(\tau, \tau') \frac{1}{i\hbar} \langle [\Delta \dot{u}_j(\tau), \Delta u_k(\tau)] \rangle$
 $+ \frac{1}{i\hbar} \langle T_{\tau} \Delta \ddot{u}_j(\tau) \Delta u_k(\tau') \rangle$

$$[\Delta \dot{u}_j(\tau), \Delta u_k(\tau)] = [\dot{u}_j(\tau) - \langle \dot{u}_j(\tau) \rangle, u_k(\tau) - \langle u_k(\tau) \rangle]$$

since $\langle u \rangle$ and $\langle \dot{u} \rangle$ are ordinary numbers, it commutes with the operators so $[\Delta \dot{u}_j(\tau), \Delta u_k(\tau)] = [\dot{u}_j(\tau), u_k(\tau)]$

$$= i\hbar \delta_{jk}$$

we note $\dot{u}_j(\tau) \equiv p_j(\tau)$

this gives the fine term on the righthand side as $I \delta(\tau, \tau')$.

The ^{Heisenberg} equation for u is given on page 315, i.e.,

$$\frac{d^2 u}{d\tau^2} = \ddot{u} = -Ku - c^+ M c - f \quad \text{all at time } \tau$$

($\frac{d}{d\tau}$ and $\frac{d}{dt}$ are the same)

so $\Delta \ddot{u} = \ddot{u} - \langle \ddot{u} \rangle$

Here u is in the full Heisenberg picture, $\langle \rangle = \text{Tr}(\rho \dots)$ is in dept of time.

$$= -Ku - c^+ M c - f + K\langle u \rangle + \langle c^+ M c \rangle + f$$

all at time τ . f is a scalar, not operator so $\langle f \rangle = f$

we get

$$\frac{\partial^2}{\partial \tau^2} D_{jk}(\tau, \tau') = -\delta_{jk} \delta(\tau, \tau') + \frac{1}{i\hbar} \langle T_{\tau} \left[-\sum_s K_{js} u_s(\tau) - \sum_{\ell \ell'} c_{\ell}^+ M_{\ell \ell'}^j c_{\ell'}(\tau) \right] \Delta u_k(\tau') \rangle$$

$$+ \frac{1}{i\hbar} \langle T_{\tau} \left[\sum_s K_{js} \langle u_s(\tau) \rangle + \sum_{\ell \ell'} c_{\ell}^+(\tau) M_{\ell \ell'}^j c_{\ell'}(\tau) \right] \Delta u_k(\tau') \rangle$$

$$\underbrace{\hspace{10em}}_{\langle \ddot{u}_j(\tau) + f_j(\tau) \rangle}$$

write $\Delta \ddot{u} = -K(u - \langle u \rangle) - c^\dagger M c + \langle c^\dagger M c \rangle$

we get

$$\begin{aligned} \frac{\partial^2}{\partial \tau^2} D_{jk}(\tau, \tau') &= \sum_{js} \delta_{jk} \delta(\tau, \tau') + \frac{1}{i\hbar} \langle T_\tau [-K_{js} \Delta u_s(\tau)] \Delta u_k(\tau') \rangle \\ &+ \frac{1}{i\hbar} \sum_{\ell \ell'} \langle T_\tau (-c_\ell^\dagger(\tau) M_{\ell \ell'}^j c_{\ell'}(\tau)) \Delta u_k(\tau') \rangle \\ &+ \frac{1}{i\hbar} \sum_{\ell \ell'} \langle T_\tau c_\ell^\dagger(\tau) M_{\ell \ell'}^j c_{\ell'}(\tau) \rangle \langle \Delta u_k(\tau') \rangle = 0 \end{aligned}$$

The last term is zero since $\langle \Delta u_k(\tau') \rangle = \langle u_k(\tau') - \langle u_k(\tau') \rangle \rangle$

we write

$$\begin{aligned} \frac{\partial^2}{\partial \tau^2} D_{jk}(\tau, \tau') &= \sum_{js} \delta_{jk} \delta(\tau, \tau') - \sum_s K_{js} D_{sk}(\tau, \tau') \\ &- \frac{1}{i\hbar} \sum_{\ell \ell'} \langle T_\tau M_{\ell \ell'}^j c_\ell^\dagger(\tau) c_{\ell'}(\tau) u_k(\tau') \rangle + \frac{1}{i\hbar} \sum_{\ell \ell'} M_{\ell \ell'}^j \langle T_\tau c_\ell^\dagger(\tau) c_{\ell'}(\tau) \rangle \times \langle u_k(\tau') \rangle \end{aligned}$$

Using the result on page 323, i.e.

$$i\hbar \frac{\delta G(1,2)}{\delta f(3)} = \frac{i}{\hbar} \langle T(c(1) c^\dagger(2) u(3)) \rangle + \dots, G(1,2) \langle u(3) \rangle$$

or take $1 \equiv (\tau, \ell)$, $2 \equiv (\tau^+, \ell')$, $3 \equiv (\tau', k)$

We can write the $c^\dagger c u$ term as

$$\begin{aligned} &+ \frac{1}{i\hbar} \sum_{\ell \ell'} M_{\ell \ell'}^j \langle T_\tau c_\ell(\tau) c_{\ell'}^\dagger(\tau^+) u_k(\tau') \rangle \\ \text{change sign here} &= - \sum_{\ell \ell'} M_{\ell \ell'}^j \left[i\hbar \frac{\delta G(\tau, \tau^+)}{\delta f(\tau')_k} - G(\tau, \tau^+) \langle u(\tau') \rangle_k \right] \end{aligned}$$

note the last term in the equation of motion

$$\text{for } D \text{ is } \frac{1}{i\hbar} \sum_{\ell \ell'} M_{\ell \ell'}^j \langle T_\tau c_\ell^\dagger(\tau) c_{\ell'}(\tau) \rangle = -\frac{1}{i\hbar} \sum_{\ell \ell'} M_{\ell \ell'}^j \langle T_\tau c_\ell(\tau) c_{\ell'}^\dagger(\tau^+) \rangle$$

$$= - \sum_{\ell \ell'} M_{\ell \ell'}^j G(\tau, \tau^+)$$

extra term is $\frac{\delta G}{\delta f}$. So we get

$$\frac{\partial^2}{\partial \tau^2} D(\tau, \tau') = -I \delta(\tau, \tau') - K D(\tau, \tau') - i\hbar \text{Tr}_e \left(M \otimes \frac{\delta G(\tau, \tau')}{\delta f(\tau')} \right)$$

This agrees with that on page 327.

Unlike the equation for u , there is no linear driven term $f(\tau)$!

outer product of two vectors

Having worked out the equation of motion for G and D , the electron & phonon Green's functions, expressing the nonlinear terms in terms of functional derivative $\frac{\delta G}{\delta f(x)}$. This step is the key insight of Hedin, the rest of steps are merely some algebraic manipulation and definition. The step parallel to what we have done for the electron Coulomb interactions.

First, we make a change of variational independent variable $f(\tau) \rightarrow f(1)$ ($1 \equiv (\tau, k)$) to $\langle u_k(\tau) \rangle = \langle u(1) \rangle$

This is done by the linear response page 315 - 321

$$\frac{\delta \langle u(1) \rangle}{\delta f(2)} = -D(1, 2)$$

Usually in linear response theory, the right-hand side is evaluated at $f \equiv 0$. However, we might also need high order derivatives, such as

$$\frac{\delta^2 \langle u(1) \rangle}{\delta f(1) \delta f(2)}$$

so that right-hand side can still be a functional of f as well (which should be correct). The precise mapping from f to $\langle u \rangle$ is (which could be nonlinear if D still depends on f)

$$\langle u \rangle_f - \underbrace{\langle u \rangle_{f=0}}_{\text{constant}} = - \int D(1, 2) f(2) d2$$

Having specified how we do change of variables from f to $\langle u \rangle$, we take functional derivative with respect to $\langle u \rangle$ instead of f , $\delta \langle u \rangle = \langle u \rangle_{f+\delta f} - \langle u \rangle_f$. $\delta \langle u(z) \rangle$ is the infinitesimal variation of $\langle u \rangle$.

so functional derivative equation for G on page 325 because

$$\vec{M} \cdot \frac{\delta}{\delta \vec{f}(z)} = \left\langle \vec{M} \cdot \frac{\delta \langle u(z) \rangle}{\delta \vec{f}(z) \delta \langle u(z) \rangle} \right\rangle \quad d_3$$

$$= \int d^3 \sum_{\vec{q}} M_{jk}^2 (-D(3, (\tau, \vec{q}))) \frac{\delta}{\delta \langle u(z) \rangle}$$

the self energy term is now

$$-i\hbar \sum_{\vec{q}} M_{jk}^2 (-D(3, (\tau, \vec{q}))) \frac{\delta}{\delta \langle u(z) \rangle} G(\tau, \vec{z})_{sk} \quad d_3 \quad \begin{matrix} 1 \equiv (\tau, j) \\ 2 \equiv (\tau', k) \end{matrix}$$

let introduce the bare vertex function as

$$M(1, 2; 3) = M_{j_1 j_2}^{\vec{j}_3} \delta(\tau_1, \tau_2) \delta(\tau_1, \tau_3)$$

This term can be written more compactly as

$$+ i\hbar \iiint M(1, 4; 5) D(3, 5) \frac{\delta}{\delta \langle u(z) \rangle} G(4, 2) d(3, 4, 5)$$

In order to identify the self energy. of the fermion

$$\int \Sigma(1, 3) G(3, 2) d_3$$

We need a Green's function G , but here G is inside the functional derivative, δG . the standard trick, see Hedin, is

$$G^{-1} G = I$$

$$\text{so } \delta G^{-1} \cdot G + G^{-1} \delta G = 0 \rightarrow -G \delta G^{-1} G = \delta G$$

$$\text{or } \frac{\delta G(4, 2)}{\delta \langle u(z) \rangle} = \iint G(4, 6) \frac{\delta G^{-1}(6, 7)}{\delta \langle u(z) \rangle} G(7, 2) d(6, 7)$$

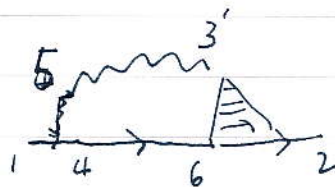
so the self energy term because, after substitute the above

$$\left(\Sigma(1,3) G(3,2) d3 = i\hbar \int d(34567) \left[M(14;5) D(3,5) \right. \right. \\ \left. \left. \times (-G(4,6) \frac{\delta G^{-1}(6,7)}{\delta \langle u(3) \rangle} G(7,2)) \right] \right.$$

re naming the dummy variables in the right hand term
 $7 \rightarrow 3, 3 \rightarrow 3'$

$$\Sigma(1,3) = -i\hbar \int d(3'456) M(14;5) D(3',5) G(4,6) \frac{\delta G^{-1}(6,3')}{\delta \langle u(3') \rangle}$$

$\downarrow$
 2 remove again $3 \rightarrow 2$



Note that, ignoring

the phonon lines (functions)

the electron line form a matrix multiplication (trace)

$$\Sigma(1 \rightarrow 2) \equiv M(14, \dots) G(4,6) \frac{\delta G^{-1}(6,2)}{\delta \dots}$$

The above diagram should be compared with that on page 299.

We define the vertex function such that to lowest order it is the same as $M(12;3)$, from the equations of motion on top of page 325, we get

$$G^{-1}(\tau, \tau') = \underbrace{i\hbar \frac{\partial}{\partial \tau} I - H}_{\text{purely symbolic}} - \underbrace{\vec{M} \cdot \langle \vec{u}(\tau) \rangle}_{\downarrow} - i\hbar \underbrace{\vec{M} \cdot \frac{\delta}{\delta f \dots}}_{\text{high order}}$$

$$G^{-1}(1,2) = \dots \dots \dots - \int M(1,2;3) \langle u(3) \rangle d3 + \text{high order}$$

$\uparrow$
 independent of $\langle \vec{u} \rangle$ $\uparrow$
 diagonal in time τ

so $\frac{\delta G^{-1}(1,2)}{\delta \langle u(3) \rangle} = -M(1,2;3)$ so we can define

vertex function

$$M(1,2;3) = - \frac{\delta G^{-1}(1,2)}{\delta \langle u(3) \rangle}$$

$\nearrow$ electron in, out
 $\uparrow$ phonon

With this definition the self energy is

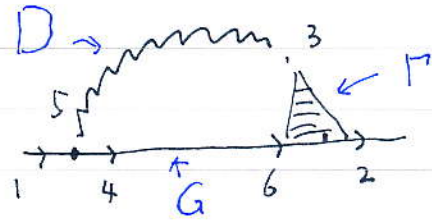
we use $D(1,2) = D(2,1)$

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$$\Sigma(1,2) = i\hbar \int d(3456) M(1,4;5) D(5,3) G(4,6) \Gamma(6,2;3)$$

$$= i\hbar \int d(3456) M(1,4;5) G(4,6) \Gamma(6,2;3) D(5,3)$$

This is identical to the electron case if $M \rightarrow (-e)$ and diagrams in the three indices of M .



If we take $\Gamma(6,2;3) \approx M(6,2;3)$ it is so-called GW method.

Now we do similar manipulation for the phonon equation of motion. The phonon self energy term is from page 327

$$\Pi D = i\hbar T_k \left(\vec{M} \otimes \frac{\delta}{\delta f(\tau')} G(\tau, \tau') \right)$$

This is symbolic only we rewrite compact notation

$$= i\hbar \sum_{ss'} M_{ss'}^j \frac{\delta G_{s's}(\tau, \tau')}{\delta f(\tau')} = i\hbar \int d(34) M(3,4;1) \frac{\delta G(4,3^+)}{\delta f(2)}$$

τ^+ mean slight later than τ .

two δ -function equal time

Do the change of independent variable for f to $\langle u \rangle$ we get

$$\frac{\delta G}{\delta f(2)} = \frac{\delta \langle u(s) \rangle}{\delta f(2)} \frac{\delta G}{\delta \langle u(s) \rangle} d5$$

$$= - \int d5 D(5,2) \frac{\delta G}{\delta \langle u(s) \rangle}$$

change again from δG to δG^{-1}

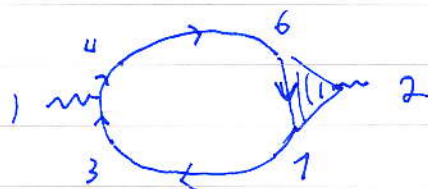
$$= - \int d5 D(5,2) \left[- G(4,6) \frac{\delta G^{-1}(6,7)}{\delta \langle u(s) \rangle} G(7,3^+) \right] d6 d7$$

$$\text{so } (\Pi D)[1,2] = i\hbar \int d(34567) \left[M(3,4;1) D(5,2) G(4,6) \right. \\ \left. - \Gamma(6,7;5) G(7,3^+) \right]$$

Compare with the definition of Π we find

$$\Pi(1,5) = (-i\hbar) \int d(3467) M(34;1) G(4,6) \Gamma(6,7;5) G(7,3^+)$$

rename $5 \rightarrow 2$



This is the polarization dig with vertex correction

This should be compared with that for electron-electron Coulomb system on page 303. (Remember $M \rightarrow (-e) \delta(13) \delta(14)$

$\delta(1,3) \rightarrow \delta(r_1, r_3)$. This demonstrated mathematical

equivalence of the pure Coulomb system & phonon system (as well as scalar photon system).

The last step is to work out a equation for the functional derivative vertex function

$$\Gamma(1,2;3) = - \frac{\delta G(1,2)}{\delta \langle u(3) \rangle}$$

Symbolically we have the Dyson equation

$$G^{-1}(1,2) = G_0^{-1}(1,2) - \underbrace{\int M(1,2;3) \langle u(3) \rangle d3}_{\text{"Hartree" term}} - \underbrace{\Sigma(1,2)}_{\text{"Fock term"}}$$

and

$$\left(i\hbar \frac{\partial}{\partial \beta} - H \right) G_0(1,2) = \delta(1,2)$$

G_0 is the Green's function with out the force perturbation F or $\langle u \rangle$. So it is independent of $\langle u \rangle$ so

$$\Gamma(1,2;3) = - \frac{\delta G^{-1}(1,2)}{\delta \langle u(3) \rangle} = M(1,2;3) + \frac{\delta \Sigma(1,2)}{\delta \langle u(3) \rangle}$$

This equation is similar to that for Coulomb system on page 305. To compute the functional derivative

$\frac{\delta \Sigma}{\delta \langle u \rangle}$ we need one more change of variables from $\langle u \rangle$ to G .

$$\frac{\delta \Sigma(1,2)}{\delta \langle u(3) \rangle} = \int d(45) \frac{\delta G(45)}{\delta \langle u(3) \rangle} \frac{\delta \Sigma(1,2)}{\delta G(4,5)} \quad \text{here, in order to}$$

have this equation to make sense, we need to view Σ as a functional of G only (with other quantities independent of $\langle u \rangle$). Since $\langle u \rangle$ is related to D so Σ cannot be a functional of D , otherwise we also need to do $\frac{\delta \Sigma}{\delta D}$. The idea is that we can express Σ in terms of G_0 and D_0 (as well as $\langle u_0 \rangle$!) since $G = G_0 + G_0 \Sigma_{\text{tot}} G$ (here Σ_{tot} include the Hartree type, Σ does not) by inverting this relation $G_0 \equiv G_0(G)$, we can eliminate G_0 in favor of G , then $\Sigma = \Sigma(G, D_0)$. Note that we don't invert D_0 , the functional derivative is in the above sense.

[This is certainly the argument used leading to the usual equation but Hedin's original 1965 paper don't have it]. D_0 is independent of f or $\langle u \rangle$. Why Σ is a function of G and D_0 and not $\Sigma(G, \langle u \rangle, D_0)$ is puzzling to me! This is the above argument is correct, then we get using the usual trick

$$\begin{aligned} \frac{\delta \Sigma(1,2)}{\delta \langle u(3) \rangle} &= \int d(45) \left[G(4,6) \frac{\delta G^{-1}(6,7)}{\delta \langle u(3) \rangle} G(7,5) \frac{\delta \Sigma(1,2)}{\delta G(4,5)} \right] d6d7 \\ &= \int d(4567) \frac{\delta \Sigma(1,2)}{\delta G(4,5)} G(4,6) \Gamma(67;3) G(7,5) \end{aligned}$$

and with $\Gamma(1,2;3) = M(12;3) + \frac{\delta \Sigma(1,2)}{\delta \langle u(3) \rangle}$ we get the last Hedin equation

$$\Gamma(12;3) = \text{diagram 1} = \text{diagram 2} + \text{diagram 3}$$

Diagram 1: A triangle with vertices 1, 2, 3. A shaded region is inside, with a cross inside. A line goes from 1 to 2, and another from 2 to 3, and a third from 3 to 1.

Diagram 2: A triangle with vertices 1, 2, 3. A shaded region is inside, with a cross inside. A line goes from 1 to 2, and another from 2 to 3, and a third from 3 to 1.

Diagram 3: A triangle with vertices 1, 2, 3. A shaded region is inside, with a cross inside. A line goes from 1 to 2, and another from 2 to 3, and a third from 3 to 1. Below the triangle is a label $\frac{\delta \Sigma(12)}{\delta G(45)}$.

We have split the electron self energy into a "Hartree" term and Fock term, only the Fock term is denoted by Σ . the hartree term is given, for the electron-phonon problem as

$$\int M(12;3) \langle u(3) \rangle d3$$

This does not look like the usual lollipop diagram



but in fact it is this diagram just as in the electron Hartree term. An important point

is that it is  with double line for electron and single line for phonon. $G \cdot D_0$.

For systems with space translationally invariance systems, such as periodic crystal lattice. This term is 0 (when $f \equiv 0$ of course) see AGD, page 79, 119. However, the argument used there cannot be applied to nanostructure without periodicity. So we keep it.

To show the hartree term is actually the Feynman diagram , we assume that we have our

final results for all the formulas and have set the driving force $f \equiv 0$. then the system is time translationally invariant, thus $\langle u(\tau) \rangle$ is independent of time τ , it is just a number (in fact, vector $\langle \vec{u} \rangle$). So in steady state without the driven force, $\langle u \rangle$ satisfies equation (see page 327)

$$0 = \frac{d^2}{d\tau^2} \langle u(\tau) \rangle = -K \langle u(\tau) \rangle + i\hbar \text{Tr}_e (\vec{M} G(\tau, \tau^+)) = 0$$

The trace here is over the electron degrees of freedom.

We can solve for $\langle u \rangle$ to get

$$\langle \vec{u} \rangle = i\hbar K^{-1} \text{Tr}(\vec{M} G(\tau, \tau^+))$$

But $D_0^r(\omega) = ((\omega + i\eta)I - K)^{-1} = -(K)^{-1}$ when $\omega = 0$

so $\langle u_j \rangle = -i\hbar \left[D_{jk}^r(\omega=0) \right] \sum_{ss'} M_{ss'}^K G_{s's}(\tau'', \tau^+)$
 result is integral of τ'' .

we get $\int M(1, 2; 3) \langle u(3) \rangle d3$

$$\Rightarrow \sum_l M_{jk}^l \delta(\tau, \tau_l) \delta(\tau, \tau_l) \langle u_l(\tau_l) \rangle d\tau_l = \sum_l M_{jk}^l \delta(\tau, \tau_l) \langle u_l(\tau_l) \rangle$$

$$\tau_1 \rightarrow \tau, \tau_2 \rightarrow \tau' \rightarrow \sum_l M_{jk}^l \delta(\tau, \tau') \langle u_l(\tau) \rangle$$

$$\rightarrow \sum_{lss'} (-i\hbar) M_{jk}^l \delta(\tau, \tau') D_{lk'}^{or}(\omega=0) M_{ss'}^{K'} G_{s's}(\tau'', \tau^+)$$

$$= (-i\hbar) \sum_{lss'} M_{jk}^l D_{lk'}^{or}(\omega=0) M_{ss'}^{K'} G_{s's}(\tau'', \tau^+) \delta(\tau, \tau')$$

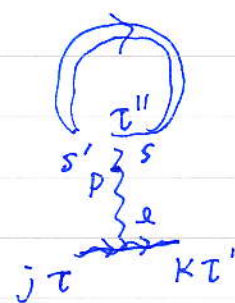
$$= (-i\hbar) \delta(\tau, \tau') \sum_{ls} M_{jk}^l \int_{-\infty}^{+\infty} D_{lp}^{or}(t) dt M_{ss'}^P G_{s's}(\tau'', \tau^+)$$

time

This is precisely the value of the diagram



→ label the point



Why only D^{or} ??

According to the Feynman rule, the value of the graph is

each vertex has only one time variable.

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$$(-i\hbar) S(\tau, \tau') \sum_{\substack{epss'}} M_{jk}^2 \int D_{ep}^0(\tau, \tau'') M_{ss'}^P \underbrace{G(\tau'', \tau''')}_{\text{2 by } \tau''} d\tau''$$

$$\text{so } \int D_{ep}^0(\tau, \tau'') d\tau'' = \int_{-\infty}^{+\infty} \dots$$

$$= \int_{-\infty}^{+\infty} D_0^{\sigma+}(\tau - \tau'') d\tau'' - \int_{-\infty}^{+\infty} D_0^{\sigma-}(\tau - \tau'') d\tau''$$

$$= \int_{-\infty}^{+\infty} [D_0^{\sigma+}(\tau) - D_0^{\sigma-}(\tau)] d\tau \quad \text{if } \sigma = + \text{ we have } D_0^{++} - D_0^{+-} = D_0^+ - D_0^- = D_0^r$$

$$\text{if } \sigma = - \text{ we also have } D_0^{-+} - D_0^{--} = D_0^+ - D_0^- = D_0^r$$

$$\equiv \int_{-\infty}^{+\infty} D_0^r(\tau) d\tau = D_0^r(\omega=0). \quad Q \in D.$$

Important lesson learned. If we do GW then the Hartree term is the wiggly line is not double line but a single line.

$$\text{Diagram: a circle with a wiggly line entering from the bottom} = D_0 G$$

So the self-consistent Born approximation used by me, Liu Jingtao, Zhang Lifu is wrong. Since we used double line



leads to double counting for some diagrams, such as



With the double line (D) we get a factor of 2 for this diagram but it should be 1.

The Madin equation derivation gives a rigorous justification why the Hartree term should be in single line Do!